

Jaewon Yoon

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RESEARCH INTEREST

My research interest lies in **developing integrated computational pipelines** to solve problems regarding **the sequence-structure-function paradigm in scale**. Specific examples would include developing computational tools applicable to **predict in vivo biomolecular interactions** and **designing molecular binders** capable to be used in disease treatment. My past research experience includes improving Folddisco, an open-source software that can search structure motifs in scale.

EDUCATION

Seoul National University

03 2021 — 02 2027

B.S. in Biological Sciences, Interdisciplinary Major of Artificial Intelligence - (GPA - 3.87/4.3)

RESEARCH EXPERIENCE

SNU Steinegger Lab (Laboratory of Machine Learning and Bioinformatics)

07 2025 - Present

Undergraduate Researcher / PI: Prof. Dr. Martin Steinegger

- Developing methods for improving metrics for outputs of **Folddisco** (a software for structural motif search).
- Developing **FrankenFold**, a hybrid protein structure predicting model, excluding the diffusion module and combining the architectures of different versions of the AlphaFold models.

SNU Laboratory of Computational Biology

12 2024 - 02 2025

Undergraduate Researcher / PI: Prof. Daehyun Baek

- Explored the applications of Deep - Learning approaches to transcriptomic data.
- Developed a deep learning model to predict RNA modifications from sequencing data produced by Nanopore sequencing.

PUBLICATIONS

Journal paper

- Kim, H., Kim, R.S., Mirdita, M., **Yoon, J.**, Steinegger, M. (2026). Structural motif search across the protein-universe with Folddisco *Nature Biotechnology*, doi: 10.1038/s41587-026-03162-9

PRESENTATIONS / POSTERS

ORAL PRESENTATION — iscb-AI

08 2026

- Oral presentation on "An Improved Metric for Searching Protein Motifs with Folddisco"

SCHOLARSHIP / AWARDS

Presidential Science Scholarship — Korea Student Aid Foundation

2021-2026

- 150 undergraduate students selected nationwide for excellency

Encouragement Prize — SNU College, Seoul National University

02 2026

- Awarded to outstanding class founders of the 2025 Student-Directed Education Programs

Dean's list — College of Natural Sciences, Seoul National University

Autumn 2021, Autumn 2025

- Awarded to outstanding undergraduate students each semester

Army Commendation Medal — Eighth Army, United States of America

08 2023

- Awarded to winners of General Paik Sun-Yup Leadership Board hosted by the Eighth Army, USFK

Korea Talented Award — Ministry of Education, Republic of Korea

11 2020

- 50 high school students and 50 university students selected nationwide

International Biology Olympiad Silver Medal

08 2020

- Scientific competition between international teams of secondary school students

EXTRACURRICULAR - ACADEMIC

Regional Student Group - Korea Core Team, ISCB

01 2026 - Present

Member, Program Committee

- Planned and organized annual symposiums, workshops, and web seminars for RSG-Korea of ISCB
- Invited multiple scholars focusing on antibody structure prediction or omics data analysis for seminars

Teaching assistant, SNU college

06 2026 - 08 2026

Undergraduate student tutor

- Assisted teaching of the course Basic Computing: First Adventures in Computing
- Assisted and advised undergraduate students in programming exercises and term projects

Deep-learning-based bioinformatics seminar - student-directed education seminar

Autumn 2025

Class founder

- Held weekly student seminars on investigating various fields and researches of computational biology
- Invited multiple scholars focusing on protein structure prediction, protein - ligand interaction prediction, and genomic data analysis for lecture
- Organized presentations and discussions regarding deep-learning based bioinformatics research

EXTRACURRICULAR - NON-ACADEMIC

SNU College of Natural Sciences Undergraduate Student Mentor (Multi-major)

2025, 2026

Undergraduate Student Mentor

- Provided advice to undergraduate students on selecting and preparing various double majors.

SNU CNS Science Volunteering Group

2022, 2024-2026

Group member / Member of Group task force - mentoring division

- Participated in mentoring and advising high school students of rural areas nationwide.
- Established and directed multiple programs to academically advise students.

SNU School of Biological Sciences Undergraduate Student Mentor

03 2025 - 02 2026

Undergraduate Student Mentor

- Advised freshman and sophomore students of the department as an undergraduate student mentor.
- Aided overall school life and provided counseling regarding career paths for fellow students.

MSC-K HQ, 19th ESC, 8th Army, USFK (Military Service)

08 2022 - 02 2024

KATUSA soldier / Discharged as army sergeant (E-5)

- Served as a Korean soldier augmented to the USFK for mandatory military service of ROK.
- Supported communication and cooperation between USFK stationed in Camp Carroll and Korean army counterparts.

REFERENCES

Prof. Dr. Martin Steinegger

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Scholar Profiles: SteineggerLab Homepage — Seoul National University - Personal Page

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Prof. Dr. Milot Mirdita

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